

Python Programming

Practical Lab Notes | All 17 Practicals

PRACTICAL 1 — Detailed Answer

Q1. Install Python and get familiar with IDLE / Anaconda / Jupyter.

What is Python?

Python is a high-level, interpreted, general-purpose programming language known for its simple and readable syntax. It supports multiple programming paradigms including procedural, object-oriented, and functional programming. Python is widely used in web development, data science, artificial intelligence, automation, and more.

Step 1 — Download & Install Python

Step	Action
1	Visit https://www.python.org/downloads/
2	Click 'Download Python 3.x.x' (latest stable version)
3	Run the installer — tick ✓ 'Add Python to PATH'
4	Click 'Install Now' and wait for completion
5	Open Command Prompt → type <code>python --version</code> to verify

Option A — IDLE (comes built-in with Python)

IDLE (Integrated Development and Learning Environment) is Python's built-in editor. It is perfect for beginners.

- Open IDLE from the Start Menu (search 'IDLE').
- Go to File → New File to open the editor.
- Type your Python code and press F5 (or Run → Run Module) to execute.
- The Shell window at the bottom shows output.

Sample IDLE session:

```
print("Hello, World!") print("Python version:", 3) x = 5 print("Value of x:", x)
```

Option B — Anaconda (recommended for Data Science)

Anaconda is a free distribution of Python that includes 250+ popular packages (NumPy, Pandas, Matplotlib, etc.) and the Conda package manager.

- Download from <https://www.anaconda.com/download>
- Run the installer (leave default settings).
- Open Anaconda Navigator from the Start Menu.
- Launch Spyder (IDE) or Jupyter Notebook from the Navigator.
- To create a new environment: `conda create -n myenv python=3.10`

Option C — Jupyter Notebook (best for interactive work)

Jupyter Notebook runs in the browser and lets you mix code, output, and notes in a single document. Each 'cell' can be run independently.

To launch Jupyter:

```
# In Anaconda Prompt or terminal: jupyter notebook
```

A browser window opens at <http://localhost:8888>. Click New → Python 3 to create a new notebook. Type code in a cell and press Shift + Enter to run it.

Comparison of Python Environments

Environment	Best For	Difficulty	Includes
IDLE	Beginners / simple scripts	Easy	Built-in with Python
Spyder	Scientific computing	Medium	With Anaconda
Jupyter	Data science / teaching	Medium	With Anaconda / pip
VS Code	Professional projects	Medium+	Free download

Your First Python Program

Open IDLE or Jupyter, type the following code, and run it:

```
# My First Python Program
print("Hello, World!")
print("Welcome to Python Programming")
name = input("Enter your name: ")
print("Hello,", name)
```

Expected Output: Hello, World! | Welcome to Python Programming | Hello, <your name>

PRACTICALS 2–17 — Questions & Hints

Q2. Display simple text messages.

■ Hint:

- Use the built-in `print()` function.
- Strings can use single quotes `'text'` or double quotes `"text"`.
- Use `\n` inside a string to print a new line.

Sample Code:

```
print("Hello World!")
print('Python is fun!')
```

Q3. Perform arithmetic operations (+, -, *, /, //, %, **).

■ Hint:

- Use variables to store numbers: `a = 10`
- Operators: `+` addition, `-` subtraction, `*` multiply, `/` divide
- Integer division: `//` Modulo (remainder): `%` Power: `**`

Sample Code:

```
a, b = 10, 3
print(a + b, a - b, a * b, a / b)
print(a // b, a % b, a ** b)
```

Q4. Find areas of Circle, Rectangle, and Triangle.

■ Hint:

- Circle: area = $\pi \times r^2 \rightarrow$ use import math then math.pi
- Rectangle: area = length $\times$ breadth
- Triangle: area = $0.5 \times$ base $\times$ height
- Take inputs using input() and convert with float()

Sample Code:

```
import math r = float(input("Radius: ")) print("Circle area:", math.pi * r * r)
```

Q5. Swap two variables.

■ Hint:

- Method 1 — Python one-liner: a, b = b, a (no temp variable needed!)
- Method 2 — Using a temp variable: temp = a; a = b; b = temp
- Print values before and after swap to verify.

Sample Code:

```
a, b = 5, 10 print("Before:", a, b) a, b = b, a print("After:", a, b)
```

Q6. Check whether a number is Even or Odd.

■ Hint:

- Use the modulo operator: number % 2
- If result is 0 $\rightarrow$ Even; otherwise $\rightarrow$ Odd.
- Use if-else statement for the decision.

Sample Code:

```
n = int(input("Enter number: ")) if n % 2 == 0: print("Even") else: print("Odd")
```

Q7. Find the largest of two and three numbers.

■ Hint:

- Two numbers: use a single if-else.
- Three numbers: use if / elif / else or the built-in max() function.
- Tip: max(a, b, c) returns the largest in one line.

Sample Code:

```
a, b, c = 4, 9, 6 if a >= b and a >= c: print(a) elif b >= c: print(b) else: print(c)
```

Q8. Check whether a number is Positive, Negative, or Zero.

■ Hint:

- Three conditions $\rightarrow$ use if / elif / else.
- Positive: $n > 0$ | Negative: $n < 0$ | Zero: $n == 0$

Sample Code:

```
n = float(input("Enter: ")) if n > 0: print("Positive") elif n < 0: print("Negative") else: print("Zero")
```

Q9. Find factorial of a number using a loop.

■ Hint:

- Factorial of $n = n \times (n-1) \times (n-2) \times \dots \times 1$ (e.g., $5! = 120$)
- Use a for loop with `range(1, n+1)` and multiply into a result variable.
- Initialize `result = 1` before the loop.

Sample Code:

```
n = int(input("Enter n: ")) result = 1
for i in range(1, n + 1): result *= i
print("Factorial:", result)
```

Q10. Print the multiplication table of a number.

■ Hint:

- Use a for loop from 1 to 10.
- Inside the loop, print: $\text{num} \times i = \text{num} * i$
- Use f-string or `format()` for neat output.

Sample Code:

```
n = int(input("Table of: "))
for i in range(1, 11): print(f"{n} x {i} = {n*i}")
```

Q11. Find sum of first N natural numbers.

■ Hint:

- Use a for loop from 1 to N, adding each value to total.
- Alternatively, use the formula: $\text{sum} = N \times (N+1) / 2$
- Built-in option: `sum(range(1, N+1))`

Sample Code:

```
n = int(input("Enter N: ")) total = sum(range(1, n + 1))
print("Sum:", total)
```

Q12. Use functions to find square and cube of a number.

■ Hint:

- Define functions using `def` keyword.
- Square function returns $n ** 2$; Cube returns $n ** 3$
- Call the functions and print results.

Sample Code:

```
def square(n): return n ** 2
def cube(n): return n ** 3
n = int(input("Number: "))
print("Square:", square(n))
print("Cube:", cube(n))
```

Q13. Find maximum of three numbers using a function.

■ Hint:

- Define a function `find_max(a, b, c)` with three parameters.
- Inside, use `if/elif/else` or simply `return max(a, b, c)`
- Call with three user-inputted values.

Sample Code:

```
def find_max(a, b, c): return max(a, b, c)
a, b, c = 3, 7, 5
print("Max:", find_max(a, b, c))
```

Q14. Create a list and display its elements.

■ Hint:

- Create a list: `my_list = [10, 20, 30, 40]`
- Access by index: `my_list[0]` (0-based)
- Loop through: `for item in my_list:` or use `enumerate()`

Sample Code:

```
my_list = [10, 20, 30, 40, 50] print("List:", my_list) for i, val in enumerate(my_list):  
print(f"[{i}] = {val}")
```

Q15. Find sum and average of elements in a list.

■ Hint:

- Use built-in `sum(list)` for total.
- Average = `sum / len(list)` — use `len()` for count.
- Round with `round(avg, 2)` for cleaner output.

Sample Code:

```
nums = [10, 20, 30, 40, 50] print("Sum:", sum(nums)) print("Avg:", sum(nums) / len(nums))
```

Q16. Find maximum and minimum in a list.

■ Hint:

- Use built-in `max(list)` and `min(list)` — simplest approach.
- Manual method: start with `mx = list[0]` and compare each element.

Sample Code:

```
nums = [3, 1, 9, 5, 7] print("Max:", max(nums)) print("Min:", min(nums))
```

Q17. Count the number of elements in a list.

■ Hint:

- Use `len(list)` to get the total count.
- To count a specific value: `list.count(value)`
- Can also use a for loop with a counter variable.

Sample Code:

```
nums = [1, 2, 3, 4, 5, 2, 2] print("Total elements:", len(nums)) print("Count of 2:",  
nums.count(2))
```